

Operational Problem

Understanding factors affecting plant throughput

Why is throughput changing?

Business Problem

- Throughput drops unpredictably
- Root cause unknown
- Internal logs insufficient

Business Impact

- Delayed production
- Poor resource allocation
- Reactive decision making

Internal Data

Pressure
Temperature
Flow Rate

Weather API

Temperature
Humidity
Conditions

Database

Maintenance
Equipment
History

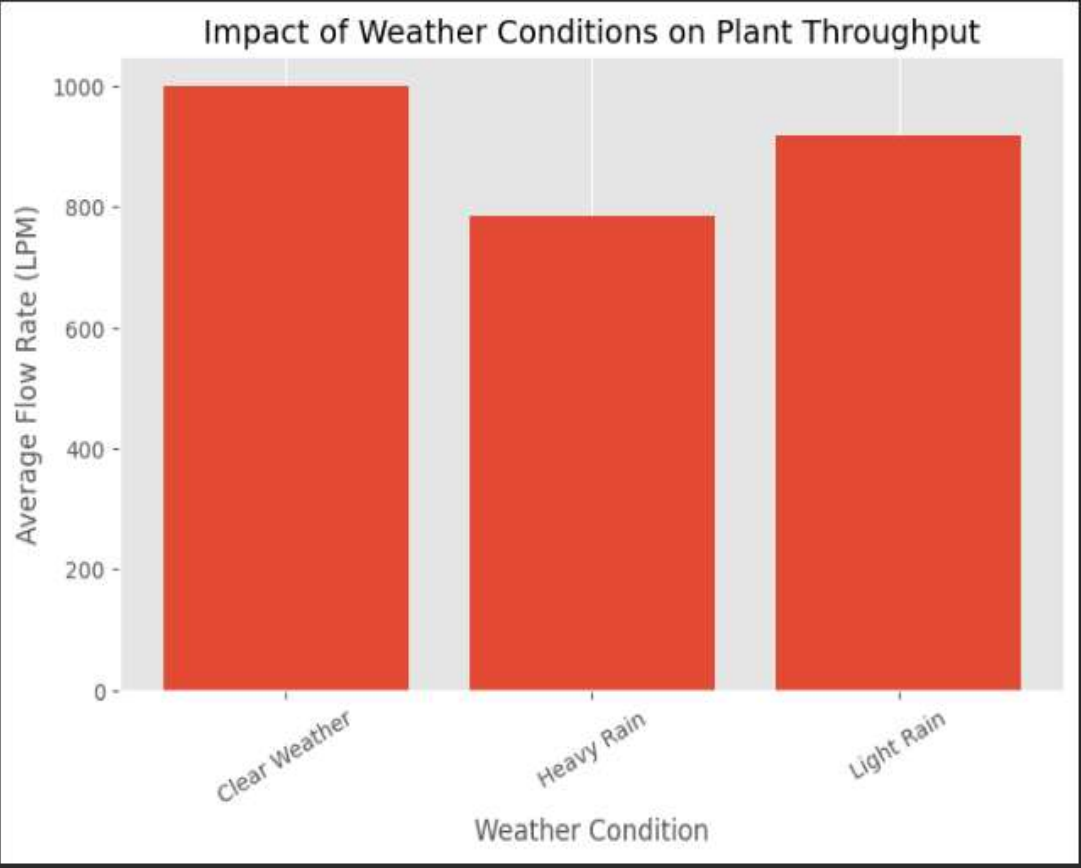
Business Question

Internal operational data explains **what** happened, but not **why** it happened. Therefore, why does weather influence throughput?



Observations & Insights

Integrated analysis from multiple data sources



Key Finding

Weather-linked throughput patterns identified. Weather conditions influenced throughput. After simulating rainfall effects, average flow rates declined during rainy periods compared with clear conditions.

Impact

External data improved visibility and schedule predictability. External data transformed isolated operational records into actionable business intelligence.



Recommendations

Using integrated data for proactive decisions

Weather Planning

Include forecasts in operations scheduling

Operations and Maintenance

Adjust during disruptions. Schedule maintenance during storms/heavy rain.

Monitoring

Build live KPI dashboards to monitor the parameters from various data sources.

Integration

Continue combining data from multiple sources to better operational forecasting and scheduling.

Better Data → Better Decisions

Business Value

- ✓ Better planning
- ✓ Reduced downtime
- ✓ Faster decisions
- ✓ Higher operational efficiency

